

Industry 4.0

Section Id :	640653122813
Section Number :	13
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	13
Number of Questions to be attempted :	13
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653328449
Question Shuffling Allowed :	No

Question Number : 247 Question Id : 6406531834717 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535954415. ✓ YES

6406535954416. ✗ NO

Sub-Section Number :	2
Sub-Section Id :	640653328450
Question Shuffling Allowed :	Yes

Question Number : 248 Question Id : 6406531834718 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct term from the Column B for the description given in Column A. Note that one to-one mapping in the table is not necessary.

Column A	Column B
a.1 You can order a mobile phone on Amazon, but can collect the product from the showroom	b.1 Impulse buying
a.2 To go to the office, you require a formal pair of trousers. While searching for trousers on Myntra, you came across a leather belt that perfectly complemented your chosen trousers, prompting you to make the purchase.	b.2 Multi-channel retailing
a.3 After acquiring a new mobile phone, you considered safeguarding it from scratches by investing in a phone cover. Now, the phone and the phone-cover are---	b.3 Complementary product
	b.4 Omni-channel retailing
	b.5 Substitution product
	b.6 Inventory-dependent demand

Options :

6406535954417. ✖ a.1 : b.4, a.2. : b.3, a.3: b.1

6406535954418. ✔ a.1 : b.4, a.2. : b.1, a.3: b.3

6406535954419. ✖ a.1 : b.4, a.2. : b.1, a.3: b.5

Sub-Section Number :

3

Sub-Section Id :

640653328451

Question Shuffling Allowed :

Yes

Question Number : 249 Question Id : 6406531834719 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

In a EOQ calculation, the estimated and actual parameter values are given in Table 1. What is the error in the EOQ calculation { $\text{error} = (Q_{\text{estimated}} - Q_{\text{actual}}) / Q_{\text{actual}}$ }. (round your answer to two decimal without the percentage sign eg, enter 10.12 if your answer is 10.12 %)

Table 1		
Parameter	Estimate	Actual
Annual demand	1000 units	2000 units
Holding cost per unit per year	\$10	\$20
Ordering cost per order	\$50	\$25

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

40 to 42

Sub-Section Number :

4

Sub-Section Id :

640653328452

Question Shuffling Allowed :

Yes

Question Number : 250 Question Id : 6406531834720 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Match the following columns by selecting the correct name from Column B for the labels in column A which are depicted in the '**Figure: EOQ model**' below:

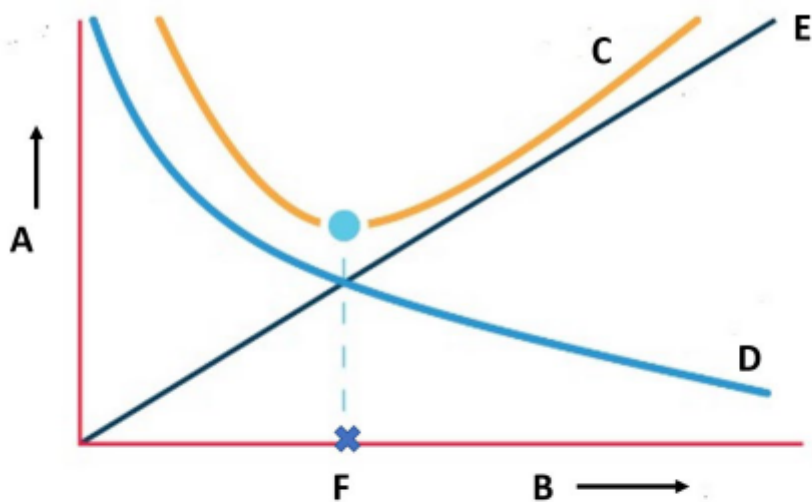


Figure: EOQ model

Column A	Column B
A	Carrying cost
B	Order cost
C	Cost
D	EOQ(Q*)
E	Quantity
F	Total cost

Options :

6406535954421. ✖ A:Cost; B:Quantity; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q*)

6406535954422. ✔ A:Cost; B:Quantity; C:Total cost; D:Order cost; E:Carrying cost; F:EOQ(Q*)

6406535954423. ✖ A:Quantity; B:Cost; C:Total Cost; D:Carrying cost; E:Order cost; F:EOQ(Q*)

6406535954424. ✖ A:Cost; B:Quantity; C:Total Cost; D:Order cost; E:Carrying cost; F:EOQ(Q*)
 6406535954425. ✖ A:Cost; B:EOQ(Q*); C:Total Cost; D:Order cost; E:Carrying cost; F:Quantity
 6406535954426. ✖ A:Total cost; B:EOQ(Q*); C:Cost; D:Order cost; E:Carrying cost; F:Quantity
 6406535954427. ✖ A:Total cost; B:Quantity; C:Cost; D:Carrying cost; E:Order cost; F: EOQ(Q*)

Sub-Section Number : 5
Sub-Section Id : 640653328453
Question Shuffling Allowed : Yes

Question Number : 251 Question Id : 6406531834721 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question

Pranjal is shopping in a departmental store. He has a list of items which he prepared at home itself. He picked a few packets of tea as part of his list. He could see coffee packets and milk-powder packets in the neighbouring shelves. He picked a few milk-powder packets, however these are not part of his list. He was searching for Good-day biscuits. He could identify only one packet in the shelf remaining. However, there were lot of Sun-fist biscuits lying in the next shelf. So he picked ten Sun-fist packets. Now map the scenarios with the items below:

Product	Scenario
P1)Tea:Coffee	S1)Complementary
P2) Tea: milk-powder packet	S2)Substitution
P3) Good-day biscuits: Sun-fist biscuit	S3)Inventory dependant demand
	S4)Impulse buying

Options :

6406535954428. ✖ P1-S1, P2-S3, P3-S4
 6406535954429. ✔ P1-S2, P2-S1, P3-S3
 6406535954430. ✖ P1-S2, P2-S1, P3-S4

Sub-Section Number : 6
Sub-Section Id : 640653328454
Question Shuffling Allowed : No

Question Id : 6406531834722 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (252 to 253)

Question Label : Comprehension

Suppose, you are responsible for material purchase in an organization. You have demand forecast of a product for next four weeks which are 3, 4, 2 and 4 units respectively. You need to purchase one type of raw material to produce the final product to meet the above demand. Consider, one unit of the final product requires one unit of this raw material. Whenever an order is placed, your organization needs to incur a cost of \$4 irrespective of the quantity ordered. Purchasing cost of the raw material is \$2 per unit. Holding cost of an item not gone for production at the end of a week is \$0.5 per week. You can order a maximum of 5 units in a week and the warehouse has

capacity to store upto 4 units of inventory in a week. Note that you need to meet the demand in every week.

Solve the above problem using dynamic programming (backward recursion) and answer the given sub-questions for this problem.

Sub questions

Question Number : 252 Question Id : 6406531834723 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What does the state variable represent?

Options :

- 6406535954431. ✖ How many units of the raw material needs to be purchased in a week
- 6406535954432. ✖ Ending inventory with the warehouse in each week
- 6406535954433. ✔ Beginning inventory with the warehouse in each week

Question Number : 253 Question Id : 6406531834724 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What does the action set represents?

Options :

- 6406535954434. ✔ How many units of the raw material needs to be purchased in a week
- 6406535954435. ✖ Ending inventory with the warehouse in each week
- 6406535954436. ✖ Beginning inventory with the warehouse in each week

Sub-Section Number :

7

Sub-Section Id :

640653328455

Question Shuffling Allowed :

Yes

Question Number : 254 Question Id : 6406531834725 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

Which of the following model corresponds to Exponential Smoothing forecasting model?

Options :

$$F_{t+1} = \alpha D_t + (1 - \alpha)F_t \text{ where } F_t, D_t \text{ represents the forecast and actual data for}$$

6406535954437. ✔ period t

6406535954438. ✖ $F_t = \mu - \theta * \epsilon_{t-1}$ where $\epsilon_{t-1} = D_{t-1} - F_{t-1}$

Sub-Section Number :

8

Sub-Section Id : 640653328456
Question Shuffling Allowed : Yes

Question Number : 255 Question Id : 6406531834726 Question Type : MCQ
Correct Marks : 1

Question Label : Multiple Choice Question

Which of the following examples can be referred as cyclic pattern?

Options :

6406535954439. ✓ Economic recession

6406535954440. ✗ Increases Ceiling Fan sales during winters

6406535954441. ✗ Increased tourist footfall towards the end of the year

Sub-Section Number : 9
Sub-Section Id : 640653328457
Question Shuffling Allowed : Yes

Question Number : 256 Question Id : 6406531834727 Question Type : MSQ
Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which are the two primary tasks of time series forecasting?

Options :

6406535954442. ✓ Identifying pattern

6406535954443. ✗ Identify variables of interest

6406535954444. ✓ Discovering relationships amongst variables of interest

Sub-Section Number : 10
Sub-Section Id : 640653328458
Question Shuffling Allowed : Yes

Question Number : 257 Question Id : 6406531834728 Question Type : MSQ
Correct Marks : 1.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the cost(s) from the following list which contribute(s) to the shortage cost:

Options :

6406535954445. ✓ Loss of customer goodwill

6406535954446. ✗ Loss due to customer return

6406535954447. ✓ Cost of losing a customer

6406535954448. ✓ Loss of profit associated with not delivering the product

Sub-Section Number : 11
Sub-Section Id : 640653328459
Question Shuffling Allowed : No

Question Id : 6406531834729 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (258 to 260)

Question Label : Comprehension

Ramu puts up a stall for Diwali Light (Diya) every year. He builds a Diya at a cost of Rs. 1 and earns Rs.2 by selling each Diya. Any Diya not sold by Diwali will be discounted to 50 Paise. From his experience, Ramu knows that the demand is Uniformly Distributed within [800, 1000] units. Then answer the given subquestions:

Sub questions

Question Number : 258 Question Id : 6406531834730 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the Critical Ratio?(round off to two decimal places)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.65 to 0.69

Question Number : 259 Question Id : 6406531834731 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the optimal order quantity that minimizes Ramu's total cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

930 to 940

Question Number : 260 Question Id : 6406531834732 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

If the cost of overstocking is less than the cost of understocking, where will the optimal order quantity lies in the interval [800, 1000] units?

Options :

6406535954451. ✖ Toward left of the mean of uniform distribution [800,1000]

6406535954452. ✖ At the mean of uniform distribution [800,1000]

6406535954453. ✓ Towards right of the mean of uniform distribution [800,1000]

Sub-Section Number : 12
Sub-Section Id : 640653328460
Question Shuffling Allowed : No

Question Id : 6406531834733 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (261 to 265)

Question Label : Comprehension

Slipline Co. requires 1500 units of plastic ribbon per month. Each plastic ribbon costs Rs. 27 in the market. Placing an order costs Rs. 150 to Slipline Co. and the inventory carrying charges work out to 20% of the unit ribbon cost per year. Calculate the given subquestions:

Sub questions

Question Number : 261 Question Id : 6406531834734 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What should be the order quantity which minimizes the total cost to Slipline Co.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1000

Question Number : 262 Question Id : 6406531834735 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the number of orders per year?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

18

Question Number : 263 Question Id : 6406531834736 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the annual order cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2700

Question Number : 264 Question Id : 6406531834737 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the average inventory in the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

500

Question Number : 265 Question Id : 6406531834738 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

What is the annual inventory carrying cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2700

Sub-Section Number : 13

Sub-Section Id : 640653328461

Question Shuffling Allowed : Yes

Question Number : 266 Question Id : 6406531834739 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct category in Column B for the costs in Column A